1. Learning setup and guarantees

1. Statistical learning objects

A batch problem declares a domain $\mathcal{X}$, labels $\mathcal{Y}$, a hypothesis class $\mathcal{H}$ of maps $h : \mathcal{X} \to \mathcal{Y}$, a loss $\ell(h(x), y)$, a distribution $\mathcal{D}$ on (x, y), and an ordered i.i.d. sample $S = ((x_i, y_i))_{i=1}^m \sim \mathcal{D}^m$.

For binary classification, $\mathcal{Y} = \{0, 1\}$ and zero-one loss is

$$\ell(h(x), y) = \mathbf{1}_{\{h(x) \neq y\}}.$$

$\mathcal{H}$ and ℓ are modeling choices; $\mathcal{D}$ is fixed but may be unknown. The sample is random before drawing and observed afterwards.

2. Population risk vs empirical risk

For $h \in \mathcal{H}$,

$$R(h) = E_{(X,Y)\sim\mathcal{D}}[\ell(h(X), Y)], \quad \hat{R}_{S(h)} = \frac{1}{m}\sum_{i=1}^{m} \ell(h(x_i), y_i).$$

$R(h)$ averages over a fresh population draw; $\hat{R}_{S(h)}$ averages over the observed sample. Under zero-one loss these are true misclassification probability and training-error fraction.

Pitfall: small $\hat{R}_{S(h)}$ is evidence about, not equality with, small $R(h)$. Keep both the hat and sample subscript.

3. ERM, realizability, and PAC

ERM returns any

$$\hat{h}_S \in \arg\min_{h\in\mathcal{H}} \hat{R}_{S(h)}.$$

Ties are allowed; the minimized quantity is sample risk.

Realizability means $\exists h^\star \in \mathcal{H} : R(h^\star) = 0$. Observed zero training error alone does not establish it.

$\mathcal{H}$ is realizably PAC learnable if one learner A and $m_{\mathcal{H}} : (0, 1)^2 \to \mathbb{N}$ exist such that, for every $(\varepsilon, \delta) \in (0, 1)^2$, every realizable $\mathcal{D}$, and every $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$,

$$P_{S\sim\mathcal{D}^m}(R(A(S)) \leq \varepsilon) \geq 1 - \delta.$$

ε is risk tolerance; δ is failure probability. Probability is over S and any learner randomness.

2. Finite-class control

1. Fixed hypothesis: Hoeffding

If h is fixed independently of the i.i.d. sample and $\ell \in [0, 1]$, then for $\gamma > 0$,

$$P\left(\left|\hat{R}_{S(h)} - R(h)\right| > \gamma\right) \leq 2\exp(-2m\gamma^2).$$

The bounded loss supplies the range; sample independence supplies the product structure.

Selection warning: $\hat{h}_S$ depends on S. Substituting it for fixed h in this statement changes an assumption. Protect data-dependent selection by controlling all candidates simultaneously.

2. Simultaneous deviation over finite $\mathcal{H}$

For fixed h, let

$$E_{h(\gamma)} = \left\{\left|\hat{R}_{S(h)} - R(h)\right| > \gamma\right\}.$$

If $\mathcal{H}$ is finite, S is i.i.d., and $\ell \in [0, 1]$, then

$$P\left(\cup_{h\in\mathcal{H}} E_{h(\gamma)}\right) \leq 2|\mathcal{H}|\exp(-2m\gamma^2).$$

Thus, with probability at least $1 - \delta$,

$$\forall h \in \mathcal{H}, \quad \left|\hat{R}_{S(h)} - R(h)\right| \leq \sqrt{\frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2m}}.$$

Equivalently, deviation at most γ is ensured by

$$m \geq \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\gamma^2}.$$

No independence between the events E_h is required.

3. What the simultaneous event gives ERM

Let $\hat{h}_S$ minimize empirical risk and let $h^\star \in \arg\min_{h\in\mathcal{H}} R(h)$. If every deviation is at most γ,

$$\begin{aligned} R\left(\hat{h}_S\right) &\leq \hat{R}_{S(\hat{h}_S)} + \gamma \\ &\leq \hat{R}_{S(h^\star)} + \gamma \\ &\leq R(h^\star) + 2\gamma. \end{aligned}$$

The middle inequality is ERM. The first and last inequalities use the same simultaneous event, which remains valid for the sample-selected hypothesis.

Pitfall: deviation tolerance γ yields an excess-risk term 2γ.

4. Sharper realizable, consistent route

With zero-one loss, call h **consistent** if $\hat{R}_{S(h)} = 0$. For fixed h with $R(h) > \varepsilon$,

$$P(h \text{ is consistent}) = (1 - R(h))^m \leq \exp(-m\varepsilon).$$

If $\mathcal{H}$ is finite, realizability holds, and ERM returns a consistent hypothesis, then

$$P\left(\exists h : R(h) > \varepsilon \text{ and } \hat{R}_{S(h)} = 0\right) \leq |\mathcal{H}|\exp(-m\varepsilon).$$

It is sufficient that

$$m \geq \frac{1}{\varepsilon}\left(\ln|\mathcal{H}| + \ln\left(\frac{1}{\delta}\right)\right)$$

to guarantee $P\left(R\left(\hat{h}_S\right) \leq \varepsilon\right) \geq 1 - \delta$.

This $\frac{1}{\varepsilon}$ rate uses realizability and consistency. It is not the two-sided $\frac{1}{\gamma^2}$ deviation result. Neither sample bound proves an ERM implementation is computationally efficient.

5. Which finite-class bound?

- Fixed h + i.i.d. bounded loss $\to$ Hoeffding.
- After selection: finite bounded-loss $\mathcal{H} \to$ simultaneous control and 2γ excess; add zero-one realizability and consistency $\to$ direct $\frac{1}{\varepsilon}$ risk rate.

Sample complexity $\neq$ computational running time.

3. Finite-sample behaviors

1. Restriction and growth function

For distinct $C = \{x_1, \ldots, x_m\}$,

$$\mathcal{H}|_C = \{(h(x_1), \ldots, h(x_m)) : h \in \mathcal{H}\} \subseteq \{0,1\}^m.$$

Count distinct label vectors, not hypotheses or parameter values. The growth function is

$$\Pi_{\mathcal{H}}(m) = \max_{C \subset \mathcal{X} : |C| = m} |\mathcal{H}|_C|.$$

For a nonempty binary class, $1 \le \Pi_{\mathcal{H}}(m) \le 2^m$. A count on one set is only a lower bound on the maximum unless every m-point set is also bounded.

For thresholds on three ordered points, infinitely many threshold values collapse to the four restrictions $000, 001, 011, 111$. The restriction count, not the parameter count, is the finite-sample object.

2. Shattering and VC dimension

$\mathcal{H}$ **shatters** C when

$$\mathcal{H}|_C = \{0,1\}^{|C|},$$

so every binary labeling of that fixed set is realized. Missing even one vector prevents shattering.

$$\mathrm{VCdim}(\mathcal{H}) = \max\{|C| : \mathcal{H} \text{ shatters } C\},$$

or ∞ if arbitrarily large finite sets are shattered.

To prove $\mathrm{VCdim}(\mathcal{H}) = d$:

- **lower bound:** exhibit one d-point set and realize all 2^d labelings;
- **upper bound:** prove every $(d+1)$-point set has a missing labeling.

Failure on one $(d+1)$-point set does not prove the upper bound.

3. Interval classifiers: VC dimension two

Let $\mathcal{I} = \left\{h_{a,b}(x) = \mathbf{1}_{\{a \le x \le b\}} : a \le b\right\}$. Two ordered points $x_1 < x_2$ are shattered: place an interval away from both, around either point, or across both to realize $00, 10, 01, 11$.

For every ordered triple $z_1 < z_2 < z_3$, labeling 101 is impossible: an interval containing z_1 and z_3 must contain z_2. Therefore

$$\mathrm{VCdim}(\mathcal{I}) = 2.$$

This is one existential two-point construction plus a universal three-point obstruction.

4. Sauer-Shelah and learning

1. Sauer-Shelah lemma

If $\mathrm{VCdim}(\mathcal{H}) \le d < \infty$, with d a nonnegative integer, then for every nonnegative integer m,

$$\Pi_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i},$$

where $\binom{m}{i} = 0$ for $i > m$. For fixed finite d, this is polynomial in m, while the unrestricted binary count is 2^m.

The lemma controls finite-sample behavior; $\mathcal{H}$ itself may be infinite.

For intervals on the five ordered points in the Tutorial, there is one all-zero vector and one for every nonempty consecutive positive block:

$$1 + 5 + 4 + 3 + 2 + 1 = 16, \quad \text{while} \quad 2^5 = 32.$$

The exact binomial sum gives a finite-m guarantee; $O(m^d)$ records only its fixed-d growth order.

2. Sauer-Shelah proof skeleton

Define $F(m,d)$ as the maximum restriction count over all binary classes with VC dimension at most d and all m-point sets. For a realized family V, delete the last coordinate: U contains all prefixes and $D \subseteq U$ those with both extensions. Then $|V| = |U| + |D|$.

Now $\mathrm{VCdim}(U) \le d$. If D shattered d old points, its two extensions would shatter $d+1$ points; thus $\mathrm{VCdim}(D) \le d-1$ and

$$F(m,d) \le F(m-1,d) + F(m-1,d-1).$$

With $F(0,d) = F(m,0) = 1$, induction closes using Pascal's identity

$$\binom{m-1}{i} + \binom{m-1}{i-1} = \binom{m}{i}.$$

3. Finite VC dimension: current conclusion

For binary zero-one classification, $d = \mathrm{VCdim}(\mathcal{H}) < \infty$ implies distribution-free realizable PAC learnability. A standard ERM upper bound has order

$$m_{\mathcal{H}}(\varepsilon,\delta) = O\left(\frac{d\ln(\frac{1}{\varepsilon}) + \ln(\frac{1}{\delta})}{\varepsilon}\right).$$

Big-O hides a constant, so this is not an exact certificate or runtime claim. Because restrictions depend on the random sample, do not blindly substitute $\Pi_{\mathcal{H}}(m)$ for $|\mathcal{H}|$ in the finite-class union bound. Agnostic equivalence and symmetrization remain Chapter 3 material.